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SITAO MIN, Chengdu University of Information Technology, Chengdu, Sichuan, China

SHIBIN ZHANG, Chengdu University of Information Technology, Chengdu, Sichuan, China

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Sitao Min
School of Cybersecurity (Xin Gu Industrial College)
Chengdu University of Information Technology
Chengdu, Sichuan, China
Advanced Cryptography and System Security Key Laboratory of Sichuan Province
Chengdu, Sichuan, China
syee777@163.com

Shibin Zhang*
School of Cybersecurity (Xin Gu Industrial College)
Chengdu University of Information Technology
Chengdu, Sichuan, China
Advanced Cryptography and System Security Key Laboratory of Sichuan Province
Chengdu, Sichuan, China
cuitzsb@cuit.edu.cn

Abstract

This paper proposes a Quantum Fuzzy Support Vector Machine (QFSVM) model by integrating fuzzy set theory, quantum computing, and Support Vector Machines (SVM). A fuzzy membership function based on the K-nearest neighbor algorithm is introduced to reduce noise impact and differentiate sample importance. Quantum kernel methods replace traditional kernels to leverage quantum advantages. For stock market forecasting, the model incorporates stock factors and processes political and social news using FinBERT to extract news-related features. The QFSVM constructs a unique feature space, capturing complex nonlinear patterns in the data. Experimental results show that QFSVM outperforms SVM in accuracy, F1 score, and precision, achieving 76.55% accuracy—a 7.72% improvement over SVM—and surpassing state-of-the-art models. Ablation studies confirm the necessity of both quantum kernels and fuzzy membership functions. This research represents an innovative fusion of quantum computing, fuzzy set theory, and SVM in finance, advancing quantum machine learning and stock market prediction while offering new opportunities and challenges for future exploration.

CCS Concepts

• **Computing methodologies** → Machine learning; Machine learning algorithms; • **Hardware** → Emerging technologies; Quantum technologies; Quantum computation.

Keywords

Support vector machine, Quantum kernel method, Fuzzy membership function, Stock market forecasting

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*Corresponding author

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1 INTRODUCTION

Stock market forecasting is a complex and fascinating field that involves a variety of statistical and machine learning methods to analyze and predict future movements in the stock market. Such forecasts are not only based on historical trading data, such as stock prices and volumes, but may also involve macroeconomic indicators, industry trends, company earnings reports, and political and social news, among other factors. With the development of artificial intelligence and big data analytics, the tools and methods of stock market analysis have undergone significant changes. Today, investors and analysts can use sophisticated algorithms and models to identify market trends and find investment opportunities. However, despite advances in technology, uncertainty in the stock market remains extremely high, which makes accurately predicting the stock market still a very challenging task.

Researchers have proposed several capable and sophisticated prediction schemes [1, 2]. We focus on support vector machines [3–6], because the SVM improves the generalization ability of the model by maximizing the width of the decision boundary. In stock market forecasting, this can help the model better distinguish between different market states and reduce the risk of overfitting. Stock data is highly nonlinear and SVM can find patterns and trends in the data by using kernel functions. Nevertheless, these approaches have drawbacks in handling massive volumes of historical financial data, intricate market dynamics, and quickly shifting circumstances. Utilizing quantum features, quantum computing has become a potentially breakthrough instrument in the area. Several researchers are currently exploring the application of quantum algorithms and their potential impact on stock market forecasting. Given its ability to solve issues at exponential speeds, quantum algorithms are a great fit for financial analysis [7–9]. However, none of these methods take into account the fuzzy and uncertain character of financial information. By introducing fuzzy logic, the uncertainty and noise in the data are better handled, outliers and overlapping classes in the data are handled more flexibly, and the adaptability of the model to complex data is improved. These features make it particularly effective in dealing with the uncertainty and fuzziness of trend forecasting in the stock market [10–13].

In this paper, SVM and fuzzy set theory are combined to deal with the problem of fuzzy stock information. A membership function

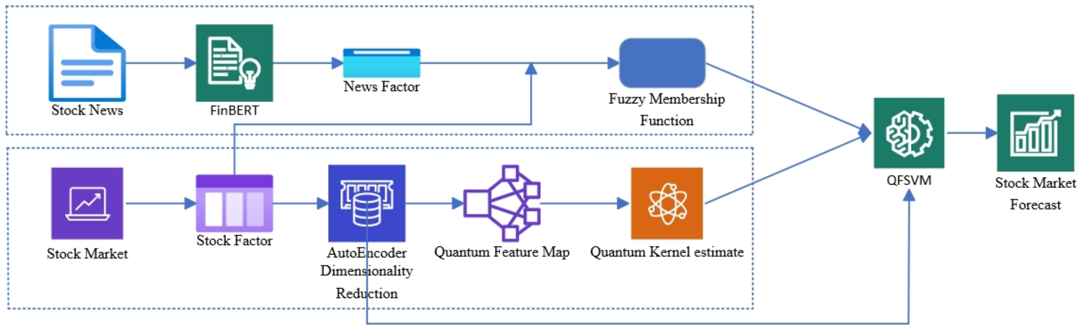

Figure 1: Overall framework.

based on the K-nearest neighbor algorithm is proposed. Using the FinBERT model [14] to extract stock news factors, fully understand news information, and combine it with stock factors. Give every training sample a fuzzy membership degree such that the relative relevance of each sample differs. The fuzzy degree of noise points and outliers is low, which in turn reduces the negative impact of noise points and outliers on the construction of the classification hyperplane. An innovative integration of support vector machines and quantum computing, using quantum kernel methods to replace traditional kernel functions to solve complex financial challenges with greater efficiency and accuracy. Finally, the QFSVM model is proposed to forecast the stock market. In the experiments, QFSVM improves the accuracy by 7.72% compared to the SVM and is superior to the state-of-the-art model. The overall framework is shown in Figure 1.

2 RELATED WORK

2.1 Fuzzy Support Vector Machines (FSVM)

An improvement on the classic SVM, the FSVM uses fuzzy logic to handle noise and uncertainty in training data. FSVM introduces fuzzy membership function, which assigns a membership value to each sample point, so that the weight of each sample point is no longer equal, and the negative impact of noise points on the classification hyperplane is less. Currently, FSVM algorithms are used in smart greenhouse [15], teaching quality evaluation [16], network traffic combination prediction [17], and many other areas have shown excellent generalization performance. Although the FSVM algorithm can handle some information with fuzzy and uncertain, it cannot handle big data effectively.

2.2 Quantum Support Vector Machines (QSVM)

QSVM is an innovative integration of classical SVM and quantum computing. Using quantum computing to address the inefficiency and high complexity of SVM algorithms in solving large amounts of data. Rebentrost et al. [18] proposed a QSVM, which introduced the least square method to transform the optimization problem of SVM into solving linear equations, and solved it by quantum HHL algorithm, which changes the time complexity from O(poly(NM)) to O(log(NM)), achieving exponential speedup. This quantum algorithms employ quantum bits with only logarithmic dataset sizes, assuming that classical data is represented in the amplitude of a quantum state. This begs the question of whether the quantum algorithm or the manner in which the data is given provides the benefit. Thus there is a QSVM based on the quantum kernel method [19], This approach avoids the assumption that the data is provided in the amplitude of the quantum state and requires only traditional access to the data. For some datasets, this method can provide demonstrable exponential speedups [20]. Although QSVM can process data faster, it does not take into consideration the fuzzy and uncertain character of realistic information, so it is interesting to study QSVM with fuzzification.

3 METHODS

We introduce a fuzzy membership function in the SVM. Not only relying on stock factors but also considering events about stock political and social news, the FinBERT model is fine-tuned to handle stock news. A membership function based on the K-nearest neighbor algorithm is proposed as a way to differentiate the importance of samples. Introducing quantum computing to replace traditional kernel functions with quantum kernel methods to gain quantum advantage. Finally fusing fuzzy set theory, quantum computing, and SVM, a QFSVM model is proposed to forecast the stock market.

3.1 Description of tasks

In this paper, we use return rankings to label stocks and consider three scenarios: positive, neutral, and negative. We define the stock market forecasting task as a categorization of these three cases. Following the definition of classification by Zou et al. [21] is shown in Eq. (1).

$$label = \begin{cases} positive & if\ r\ ranked\ top\ 20\% \\ neutral & if\ r\ ranked\ 40\% \sim 60\% \\ negative & if\ r\ ranked\ bottom\ 20\% \end{cases} \tag{1}$$

Where r is the rank of a stock's return in the market.

3.2 Stock News Processing

FinBERT [14] is a pre-trained natural language processing model. This model is based on the BERT architecture and can effectively understand and process finance-related textual data. The pre-training corpus used mainly contains financial and financial news, announcements of listed companies, and financial encyclopedia entries. Word-level pre-training and task-level pre-training were the two primary pre-training task types that were employed. The first kind of subtasks in word-level pretraining are next-sentence prediction and finance whole-word masking. The financial news financial entity recognition challenge and the industry categorization task each required task-level pre-training.

In this paper, FinBERT is fine-tuned to handle stock news information, using the fine-tuned dataset to perform multiple rounds of training loops where, in each small batch, the input data is passed to the model, losses are computed, backpropagation is performed and the model parameters are updated. Modify the FinBERT model structure by adjusting the output layer and adding a softmax layer to get the probability of the category that the news belongs to, and pass the maximum probability value as the stock news factor into the QFSVM to get the fuzzy membership function values of the training samples. The loss function we use is the cross-entropy loss function as shown in Eq. (2).

$$H\left(p,q\right)=\sum_{i=1}^{n}p\left(x_i\right)log\frac{1}{q\left(x_i\right)}=-\sum_{i=1}^{n}p\left(x_i\right)\log q\left(x_i\right) \quad (2)$$

3.3 Membership function based on K Nearest Neighbor algorithm

This method proposed in this paper combines the K Nearest Neighbor (KNN) algorithm with fuzzy logic principles for solving uncertainty and noise in classification problems. This can be used to provide a more flexible and refined decision-making process for classification. Algorithm 1 describes the fuzzy membership function based on the k-nearest neighbor algorithm.

Algorithm 1 Membership function based on K-nearest neighbor algorithm

Input: The number of neighbors K and the sample x
Output: Fuzzy degree of membership function μ(x) for sample x
1. Calculate the distance of sample x from all data in the training set.
2. Find the K nearest neighbor to sample x.
3. For sample x_i belonging to class C_j , the degree of membership is calculated by:

$$\mu\left(x\right)=\frac{\sum_{i=1}^{K}w_i\cdot\delta\left(y_i,C_j\right)}{\sum_{i=1}^{K}w_i} (3)$$

Where y_i is the class of the ith neighbor. $\delta(y_i,C_j)$ is an indicator function. If $y_i=C_j$, then $\delta(y_i,C_j)$=1, otherwise $\delta(y_i,C_j)$=0. w_i represents the news weight, obtained from the FinBert model.

As can be seen from the above membership function, we not only consider the relationship between the sample point and its surrounding neighbors but also incorporate the news weights that each neighbor possesses on its own. When the sample point and the nearby neighbors belong to different classes or the weight of the neighbors with the same attribute is small, then the fuzzy degree of membership of the sample point will be small, or even tend to 0, which is most likely to be a noise point.

3.4 Quantum kernel methods

Quantum kernel method [19] is a cutting-edge technology that combines quantum computing and machine learning kernel methods. Its core idea is to use the properties of quantum computing, specifically quantum superposition and entanglement, to process and analyze data. The quantum kernel approach first requires mapping the input data into the quantum state space. This process is typically achieved through quantum circuits, where Hadamard gates and parameterized rotation gates are used to create superpositions and rotations of quantum states. The final quantum kernel function computes the inner product of the input data through a quantum circuit, thereby quantifying the similarity between the data points. On a quantum computer, the probability that the output is all zero can be estimated by a specific quantum kernel estimation circuit, and thus the kernel function can be calculated.

Quantum kernel method can effectively process large-scale high-dimensional data through quantum computing, and improve the performance of machine learning models through the feature mapping of quantum state space. At the same time, all the features of the data can be calculated in parallel, so as to achieve the purpose of acceleration. This is due to the superposition nature of quantum states, which makes quantum algorithms potentially exponentially increase in processing speed. And quantum nuclei can capture complex nonlinear relationships through superposition and entanglement of quantum states, which is difficult for traditional kernel functions to do.

Quantum kernel methods are based on quantum computational properties that map data points from primitive space to quantum Hilbert space. As shown in Figure 2, mapping classical data x to a quantum state $|\phi(x)\rangle$ using quantum feature map. The inner product of two quantum states in Hilbert space is known as the quantum kernel function K, as shown in Eq. (4).

$$\begin{aligned}K\left(x_i,x_j\right)&=Tr\left[\left|\phi\left(x_i\right)\right\rangle\left\langle\phi\left(x_i\right)\right|\cdot\left|\phi\left(x_j\right)\right\rangle\left\langle\phi\left(x_j\right)\right|\right]\\&=\left|\left\langle\phi\left(x_i\right)\middle|\phi\left(x_j\right)\right\rangle\right|^2\end{aligned} \quad (4)$$

The feature mapping ZZFeatureMap we used is shown in Eq. (5).

$$\mathcal{U}_{\Phi}\left(\vec{x}\right)=U_{\Phi(\vec{x})}HI^{\otimes n}U_{\Phi(\vec{x})}H^{\otimes n} \quad (5)$$

where n denotes the number of quantum bits (feature dimension),H is Hadamard door, $U_{\Phi(\vec{x})}$ is a diagonal gate based on Pauli-Z, as shown in Eq. (6).

$$U_{\Phi(\vec{x})}=\exp\left(i\sum_{S\subseteq[n]}\phi_S\left(\vec{x}\right)\prod_{i\in S}Z_i\right) \quad (6)$$

The quantum circuit ZZFeatureMap is shown in Figure 3. The specific expression is shown in Eqs. (7) and (8).

$$\mathcal{U}_{\Phi}\left(\vec{x}\right)=U_{\Phi(\vec{x})}H^{\otimes 3}U_{\Phi(\vec{x})}H^{\otimes 3} \quad (7)$$

$$\begin{aligned}U_{\Phi(\vec{x})}=exp(&i\phi_1\left(x\right)Z+i\phi_2\left(x\right)Z+i\phi_3\left(x\right)Z+i\phi_{1,2}\left(x\right)\\&ZZ+i\phi_{1,3}\left(x\right)ZZ+i\phi_{2,3}\left(x\right)ZZ\end{aligned} \quad (8)$$

In Eq. (8), $\phi_i(x)=x_i,\phi_{i,j}(x)=(\pi-x_i)(\pi-x_j)$.

As shown in Figure 4. The quantum kernel function is estimated by the process of quantum kernel estimation (QKE). The initial state of the quantum circuit is $|0\rangle^n$, after the quantum line$U^{\dagger}(x_i)U(x_j)$,

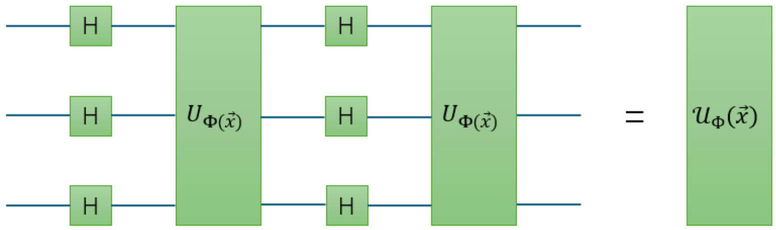

Figure 2: Quantum feature map.

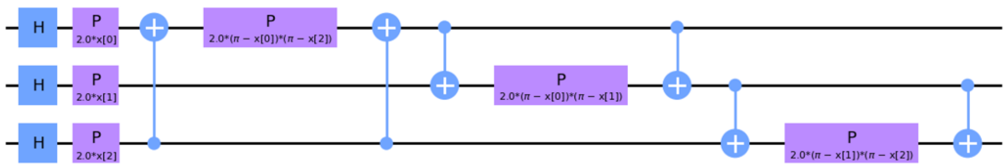

Figure 3: ZZFeatureMap quantum circuit.

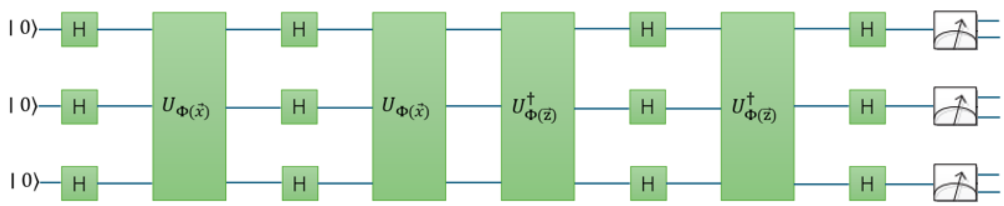

Figure 4: Estimation of quantum kernel function using QKE.

the final measurement output is the probability of 0^n as an estimate of the quantum kernel function.

3.5 Data dimensionality reduction

Dimensionality reduction is effective in classification, data visualization, and storage of high-dimensional data. In this paper, we use AutoEncoder dimensionality reduction [22], which is a nonlinear dimensionality reduction algorithm. The algorithm consists of two primary components: an encoder and a decoder. The encoder compresses the input data into a lower-dimensional representation, while the decoder maps this lower-dimensional representation back to the original data space. The primary objective of the AutoEncoder model is to reproduce the input data as accurately as possible while limiting the dimensionality of the encoding. This process forces the model to learn essential features from the data. The difference between the original data and the reconstructed output is termed the reconstruction error, and the optimization goal is to minimize this error. By doing so, the model effectively captures the most significant characteristics of the input data. The loss function used is the mean square error (MSE) as shown in Eq. (9).

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{9}$$

3.6 Quantum fuzzy support vector machines

The QFSVM proposed in this paper replaces the traditional kernel function with a quantum kernel approach and introduces a membership function. Fuzzy set theory allows elements to express the degree to which they belong to a set with a value between 0 and 1. This allows fuzzy sets to deal with fuzzy and uncertainty problems that are common in the real world. The data in QFSVM is as follows:

$$S = \{(x_i, y_i, \mu_i)\}_{i=1}^{l} = (x_1, y_1, \mu_1), (x_2, y_2, \mu_2), \cdots, (x_l, y_l, \mu_l) \tag{10}$$

Where $x_i \in R^d$ denotes the features of the training data,$y_i \in \{+1, -1\}$ denotes the class labels of the training samples. $\mu_i \in [0, 1]$ is the extent to which the training sample x_i belongs to y_i.

Like the SVM, the goal of the QFSVM algorithm is to find an optimal separating hyperplane $w^T \phi(x) + b = 0$ maximizing the interval between the two types of data. When the samples are nonlinearly differentiable, The following optimization issue is thought to be solved by locating the ideal hyperplane in a high-dimensional eigenspace:

$$\begin{cases} \min \quad \frac{1}{2}\|w\|^2 + C\sum\limits_{i=1}^{l} \mu_i \xi_i, \\ s.t. \quad y_i \left(w^T \phi(x_i) + b \right) \geq 1 - \xi_i, \\ \qquad \xi_i \geq 0, \ i = 1, 2, \ldots, l \end{cases} \tag{11}$$

The trade-off between increasing the margins and decreasing the training error term is managed by the regularization parameter C > 0, and $\xi = (\xi_1, \ldots, \xi_l)^T$ is the slack variable.

The Lagrangian function is used to convert the unconstrained problem in order to solve the optimization challenge mentioned above:

$$
\begin{aligned}
L(w, b, \alpha, \beta, \xi) = \tfrac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\mu_i\xi_i + \\
\sum_{i=1}^{l}\alpha_i\left(1 - \xi_i - y_i\left(w^T\phi(x_i) + b\right)\right) - \sum_{i=1}^{l}\beta_i\xi_i
\end{aligned}
\tag{12}
$$

where $\alpha_i \geq 0, \beta_i \geq 0$ are the corresponding Lagrange multipliers for each training sample.

A partial derivation of w 、b 、ξ_i in Eq. (12) can be obtained by making the result zero:

$$
\begin{cases}
\frac{\partial L(w,b,\alpha,\beta,\xi)}{\partial w} = w - \sum_{i=1}^{l}\alpha_i y_i \phi(x_i) = 0 \\
\frac{\partial L(w,b,\alpha,\beta,\xi)}{\partial b} = \quad -\sum_{i=1}^{l}\alpha_i y_i = 0 \\
\frac{\partial L(w,b,\alpha,\beta,\xi)}{\partial \xi_i} = \quad \mu_i C - \alpha_i - \beta_i = 0
\end{cases}
\tag{13}
$$

Substituting Eq. (13) into Eq. (11) and taking the maximum value for α_i transforms the optimization problem into a quadratic programming problem as follows.

$$
\begin{cases}
\max_{\alpha_i} \sum_{i=1}^{l}\alpha_i - \frac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l}\alpha_i\alpha_j y_i y_j K(x_i, x_j) \\
s.t. \sum_{i=1}^{l}\alpha_i y_i = 0,\ 0 \leq \alpha_i \leq s_i C,\ i = 1, 2, \ldots, l
\end{cases}
\tag{14}
$$

where $K(x_i, x_j) = \phi(x_i)^T\phi(x_j)$ is the quantum kernel function.If $\alpha_i = 0$, the corresponding samples are non-support vectors;$0 < \alpha_i \leq s_i C$ the corresponding samples are support vectors, which play a decisive role in the decision hyperplane.

Solve Eq. (14) to find $\alpha^* = (\alpha_1, \alpha_2, \cdots, \alpha_l)^T$ and the final decision function is:

$$
f(x) = sgn\left(w^T\phi(x) + b\right) \tag{15}
$$

Where $w^* = \sum_{i=1}^{l}\alpha_i^* y_i(x \cdot x_i), b^* = y_j - \sum_{i=1}^{l}\alpha_i^* y_i(x_i, x), i \in \{i | 0 < \alpha_i \leq s_i C\}$

For the sample x to be classified, the class to which it belongs is obtained by bringing it into equation (15).

3.7 Integration of classification results

Although support vector machines were first used for binary classification tasks, there are several ways in which they might be expanded to handle multiclassification tasks. In this research, we design three QFSVMs, each of which solves a subproblem between two classes, using a one-to-one strategy for our three-classification assignment. For prediction, each classifier gives a class prediction, and the final class is determined by voting; whichever class receives the most votes is predicted to be the final class.

4 EXPERIMENTATION

4.1 Data set

In this paper, we use the Astock [23] dataset proposed by Li and Shen et al. This contains stock news and stock factors for the period 2018-07-01 to 2021-11-30. Stock Factors contains 11 overbought and oversold technical indicators, 7 momentum reversal technical indicators, 4 trend technical indicators, 2 volatility technical indicators, 2 volume technical indicators, 9 liquidity factors, 6 volatility factors, 2 momentum factors, 6 value factors, and 1 leverage factor. In this paper, the segmentation strategy we use is as follows [23]: This includes 2018-07-01 through 2020-7-01 for training, 2020-7-01 through 2020-12-31 for validation, and 2021-01-01 through 2021-11-30 for testing.

4.2 Experimental platform environment

For quantum experiments, we use IBM's qiskit V1.1.0 for local simulations. The experiments were run on a desktop computer with a 13th Gen Intel(R) Core(TM) i5-13600KF CPU at 3.50 GHz, 32 GB of RAM, and an NVIDIA GeForce RTX 4070 super GPU.

4.3 Experimental details

In this paper, the FinBERT model uses a cross-entropy loss function, the optimizer uses AdamW, the learning rate is set to 1e-5, and the batch size is set to 48. The loss function used by AutoEncoder for dimensionality reduction is MSE. The number of neighbors K in the fuzzy membership function based on KNN is set to 5. The regularization parameter C in QFSVM is set to 1.

4.4 Comparison of experimental results

Our proposed QFSVM model is applied to the Astock dataset, and to evaluate the performance of our proposed model, performance metrics (accuracy, F1 score, precision) are used to compare it with current state-of-the-art models. The experimental results are shown in Table 1.

5 INTERPRETABILITY ANALYSIS

We use LIME to analyze the interpretability of our model and judge the contribution degree of three features to the final judgment of the model after AtuoEncoder dimensionality reduction. LIME is a locally explainable model-independent interpretation method that can explain any decision of any model. The core idea of LIME is to explain the decision process of the original model by building a simplified model. The main steps include: randomly extracting neighborhood data from the original data set, training a simplified model on the neighborhood data, and using the simplified model to explain the decision of the original model. LIME works by generating a set of "virtual samples" around a particular sample that are similar to the original data, and using these virtual samples to train an interpretive model. Then, by analyzing this explanatory model, an interpretation of the prediction for that sample can be obtained. The results are shown in Figure 5 and Figure 6.

6 CONCLUSION

This paper describes a concrete implementation of QFSVM and fully validates its performance on a stock market prediction task.

Table 1: Performance comparison of different models.

Model	Reference	Resource	Acc	F1	Pre
StockNet [24]	ACL 2018	News	46.72	44.44	47.65
HAN Stock [25]	WSDM 2018	News	57.35	56.61	58.41
Bert Chinese [26]	ACL 2019	News	59.11	58.99	59.07
ERNIE-SKEP [27]	ACL 2020	News	60.66	60.66	61.85
XLNET Chinese [28]	EMNLP 2020	News	61.14	61.19	61.60
RoBERTa WWM Ext [28]	EMNLP 2020	News+Factors	62.49	62.54	62.59
Self-supervised SRLP [21]	FinNLP 2022	News+Factors	66.89	66.92	66.92
SRL &SDPG&Factors [23]	WWW 2024	News+Factors	75.40	75.12	75.42
SVM	Ours	News+Factors	68.83	68.22	68.91
QSVM		News+Factors	74.13	73.79	73.98
FSVM		News+Factors	74.12	73.92	74.00
QFSVM		News+Factors	76.55	76.35	76.36

Figure 5: LIME result 1.

Figure 6: LIME result 2.

The experimental results show that the QFSVM proposed in this paper is better than the baseline model SVM in terms of accuracy, F1 score, and precision, with an accuracy of 76.55%, which is a 7.72% improvement over SVM and outperforms the current state-of-the-art models.

Also, corresponding ablation experiments are done in this paper. QSVM introduces quantum kernel methods that not only have the speed advantage of quantum computing parallelism but are also better able to characterize the internal features of the data resulting in higher accuracy. FSVM introduces the fuzzy set theory, which is better able to deal with the fuzzy and uncertainty of stock information. A fuzzy membership is assigned to each training sample so that each sample has a different level of importance, allowing noise points and outliers to have a lower fuzzy membership, which in turn reduces the negative impact of noise points and outliers when constructing the optimal classification hyperplane. Both QSVM and FSVM show significant improvement compared to the baseline SVM, and the experimental results indicate that both the quantum kernel method as well as the fuzzification process are indispensable parts of the QFSVM model. QFSVM, which combines the quantum kernel method and fuzzy set theory, can efficiently and accurately handle the complexity and uncertainty of financial data. More importantly, it opens up a new way to deal with large amounts of data with fuzzy information and promotes the practicalization of quantum machine learning.

The encouraging results of QFSVM in stock market prediction demonstrate the significant potential of quantum computing, fuzzy theory, and support vector machines at the intersection of finance. This research provides exciting opportunities and significant challenges for future exploration. In the future, we will focus on how to utilize the results of stock market prediction to support an actual trading system that is profitable enough.

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